

ROLL NO: 250701573

Experiment no:1 ANALYZE THE TREND OF DATA

1)a.Analyze the trend of data

```
import pandas as pd

import matplotlib.pyplot as plt

data ={

    "Year":[2011,2012,2013,2014,2015,2016,2017,2018,2019,2020],

    "Job_Postings":[180,320,470,650,850,1200,1650,2200,3000,4000]

}

df=pd.DataFrame(data)

print(df)

plt.plot(df["Year"], df["Job_Postings"], marker="o")

plt.title("Trend of Data Science Job Postigs")

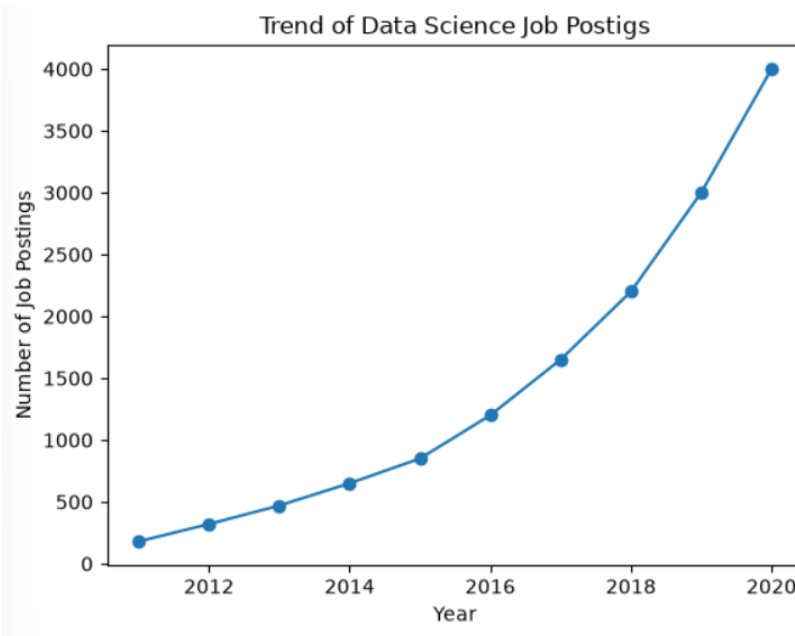
plt.xlabel("Year")

plt.ylabel("Number of Job Postings")

plt.show()
```

OUTPUT:

	Year	Job_Postings
0	2011	180
1	2012	320
2	2013	470
3	2014	650
4	2015	850
5	2016	1200
6	2017	1650
7	2018	2200
8	2019	3000
9	2020	4000



1)b. Structured.SemiStructured and UnStructured Data

```
import pandas as pd
```

```
import json
```

```
structured_data = {  
    "ID": [101, 102, 103],  
    "Name": ["Arun", "Priya", "Karthik"],  
    "Department": ["CSE", "IT", "CSE"]  
}
```

```
structured_df = pd.DataFrame(structured_data)
```

```

print("STRUCTURED DATA")

print(structured_df)

semi_structured_data = {
    "students": [
        {"ID": 101, "Name": "Arun", "Department": "CSE"},
        {"ID": 102, "Name": "Priya", "Department": "IT"},
        {"ID": 103, "Name": "Karthik", "Department": "CSE"}
    ]
}

```

```

print("\nSEMI-STRUCTURED DATA")

print(json.dumps(semi_structured_data, indent=4))

```

```

unstructured_data = """
Arun is a Computer Science student.
Priya is an Information Technology student.
Karthik is interested in Data Science.
"""

```

```

print("\nUNSTRUCTURED DATA")

print(unstructured_data)

```

OUTPUT:

```

STRUCTURED DATA
   ID  Name Department
0  101  Arun         CSE

```

```
1 102    Priya      IT
2 103    Karthik    CSE
```

SEMI-STRUCTURED DATA

```
{
  "students": [
    {
      "ID": 101,
      "Name": "Arun",
      "Department": "CSE"
    },
    {
      "ID": 102,
      "Name": "Priya",
      "Department": "IT"
    },
    {
      "ID": 103,
      "Name": "Karthik",
      "Department": "CSE"
    }
  ]
}
```

UNSTRUCTURED DATA

Arun is a Computer Science student.
Priya is an Information Technology student.
Karthik is interested in Data Science.

1)d. Encrytion and Decrytipon using Fernet

```
from cryptography.fernet import Fernet
```

```
key = Fernet.generate_key()
```

```
f = Fernet(key)
```

```
data = "My secret data"
```

```
encrypted = f.encrypt(data.encode())
```

```
decrypted = f.decrypt(encrypted).decode()
```

```
print("Original Data:", data)
```

```
print("Encrypted Data:", encrypted)
```

```
print("Decrypted Data:", decrypted)
```

OUTPUT:

Original Data: My secret data

Encrypted Data: b'gAAAAABqgcxTCMsPJb45penaT5wtQ4gkKpw8M7xo6xY-AyA1PPW5VitIoF0f4VLHxn-aBRyRLm02D3UuxZ9iQM20DAMtYbnGrQ=='

Decrypted Data: My secret data

